Failure of a pairwise four-step bound for restarted Anderson acceleration

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Exact diagonal examples refute a proposed pairwise formula for four-step residual amplification. One example has both the iteration matrix and its complement positive definite. A positive-definite one-parameter family gives an unbounded ratio of actual amplification to the proposed expression. These finite-step statements do not resolve the separate asymptotic convergence-factor conjecture.

1 The finite-step assertion

For symmetric M with $A = I - M$ nonsingular, define

$$\alpha(v) = \frac{v^T A v}{v^T A^2 v}, \qquad R(v) = M[I - \alpha(v)A]v \quad (v \neq 0), \qquad R(0) = 0.$$

The residual after four original steps is $R(R(v))$. For eigenvalues m_i, let

$$\Lambda(M) = \max_{i<j} \left(\frac{m_i m_j (m_j - m_i)}{|m_i(1-m_i)| + |m_j(1-m_j)|} \right)^2 .$$

For the formula to be defined, each pair considered must have nonzero denominator; all pairs in the examples below do. The target assertion as recovered is $\max_{v \neq 0} \|R(R(v))\|_2 \,/\, \|v\|_2 = \Lambda(M)$. The square is part of the proposed norm amplification; squared-norm comparisons require $\Lambda(M)^2$.

2 Exact examples

For $M = \operatorname{diag}(1/10, 1/2, 3/5)$ and $v = (1,1,1)^T$, direct rational substitution gives

$$\alpha(v) = 90/61, \quad R(v) = (-2, 8, 15)^T/61, \quad \alpha(R(v)) = 3140/1381,$$

$$R(R(v)) = (289, -756, 1125)^T/84241, \qquad \Lambda(M) = 1/121.$$

Consequently

$$\frac{\|R(R(v))\|_2^2}{\|v\|_2^2} = \frac{1920682}{21289638243} > \frac{1}{14641} = \Lambda(M)^2,$$

because $1920682 \cdot 14641 = 28120705162 > 21289638243$. Both M and $I - M$ are positive definite, and $\rho(M) = 3/5 < 1$.

For the simpler semidefinite $M = \mathrm{diag}(0, 1/2, 2/3)$,

$$\alpha(v) = 66/49, \quad R(v) = (0, 8, 18)^T/49, \quad \alpha(R(v)) = 35/13,$$
$$R(R(v)) = (0, -18, 16)^T/637, \qquad \Lambda(M) = 4/289.$$

Here the squared comparison is $580/1217307 > 16/83521$, certified by $48442180 > 19476912$ after cross multiplication. No optimization or tolerance is involved.

3 Unbounded underestimation

Let $M_\varepsilon = \mathrm{diag}(\varepsilon^2, 1/2, 1/2 + \varepsilon)$, $v = (1, 1, 1)^T$, with $0 < \varepsilon < 1/2$. Both M_ε and its complement are positive definite. We claim

$$\frac{\|R_\varepsilon(R_\varepsilon(v))\|_2}{\|v\|_2} = \frac{\varepsilon}{6\sqrt{6}} + O(\varepsilon^2), \qquad \Lambda(M_\varepsilon) = \frac{\varepsilon^2}{4} + O(\varepsilon^3).$$

All intermediate denominators are nonzero at $\varepsilon = 0$. Expansion gives

$$\alpha_\varepsilon(v) = \tfrac{4}{3} + \tfrac{2}{9}\varepsilon + O(\varepsilon^2).$$

For $w_\varepsilon = R_\varepsilon(v)$,

$$w_0 = (0, 1/6, 1/6)^T, \qquad w_0' = (0, -1/18, 17/18)^T.$$

With $a(\varepsilon) = (1 - \varepsilon^2, 1/2, 1/2 - \varepsilon)$, the second coefficient is

$$\beta_\varepsilon = \frac{\sum_i a_i w_{\varepsilon,i}^2}{\sum_i a_i^2 w_{\varepsilon,i}^2} = 2 + 2\varepsilon + O(\varepsilon^2).$$

Its numerator has value $1/36$ and derivative $13/108$ at zero, and denominator has value $1/72$ and derivative $5/108$. Differentiating the second residual gives

$$R_\varepsilon(w_\varepsilon) = \varepsilon(0, -1/12, 1/12)^T + O(\varepsilon^2),$$

proving the norm expansion. The pair $1/2, 1/2 + \varepsilon$ contributes

$$\left(\frac{\varepsilon(1 + 2\varepsilon)}{2 - 4\varepsilon^2}\right)^2 = \varepsilon^2/4 + O(\varepsilon^3).$$

Pairs involving ε^2 contribute $O(\varepsilon^4)$, since their denominators tend to $1/4$. Thus the former pair is maximal for sufficiently small ε, proving the second expansion. Their ratio is

$$\boxed{\frac{2}{3\sqrt{6}}\varepsilon^{-1} + O(1) \longrightarrow \infty.}$$

No fixed multiplicative constant repairs the proposed expression as a uniform four-step upper bound on positive-definite contractions.

4 Independent recurrence check and limits

The code independently executes two cycles of

$$r_1 = Mr_0, \quad d = r_1 - r_0, \quad \gamma = -r_1^T d/(d^T d), \quad r_2 = M(r_1 + \gamma d).$$

Since $d = -Ar_0$, this gives $r_2 = R(r_0)$. The exact vectors and inequalities are checked in both formulations. The continuous-family result is analytic, not established by finite sampling. A large amplification at one residual need not persist on an orbit; the asymptotic-convergence conjecture remains outside these results.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-18, *Exact four-step amplification for restarted Anderson acceleration*; O. A. Krzysik, H. De Sterck and A. Smith, *Asymptotic convergence of restarted Anderson acceleration for certain normal linear systems*, SIAM J. Sci. Comput. 47 (2025), DOI 10.1137/24M1672262, arXiv:2312.04776v4, equation (27).[1] The finite-step assertion in Section 1 is the precise mathematical target here. No assertion is made that a different formula, exponent convention, or asymptotic statement in the original paper is refuted without verifying that source.

[1] `https://arxiv.org/html/2312.04776v4`. Bibliographic details and exact target scope are retained as source leads, not independently verified during reconstruction.